MOHAMED ASRIFF

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SUMMARY

IT undergraduate seeking an IoT / Embedded Systems internship, with hands-on software development experience across 6+ academic and personal projects. Comfortable with C-family languages (Java, Python, Go) and backend/API development relevant to device-to-cloud data flows, alongside SQL/NoSQL databases suited to sensor and telemetry data storage. Strong foundation in problem solving, Linux/Git workflows, and Agile team collaboration, with a keen interest in connected devices and firmware development.

EDUCATION

IT Undergraduate **2024 - Present**
Institute of Technology, University of Moratuwa *Sri Lanka*

TECHNICAL SKILLS

IoT & Embedded Systems: ESP32, Arduino, STM32, Sensors Integration
Languages: C/C++, Python, Java, Assembly language (basic)
Backend & APIs: Spring Boot, Node.js, REST API design
Frontend: React, HTML, CSS, Figma
Databases: MySQL, PostgreSQL , relational & NoSQL data modeling
Tools & Cloud: Git/GitHub, VS Code, Arduino IDE, AWS (basic), Vercel, Azure, Canva, Postman
Other: Problem solving, Agile teamwork, project & team management, communication

PROJECTS

Autonomous Line Following Robot | Github *[C++]*

- Designed and developed an autonomous line-following robot capable of following a predefined path.
- Integrated IR sensors to detect line positions and identify path directions.
- Implemented motor control logic to adjust robot movement based on sensor readings.
- Worked with motor drivers and DC motors to achieve accurate navigation and movement.

Smart Medicine Container| Github *[C++]*

- Developed an RFID-based smart medicine container with electronic locking functionality.
- Integrated RFID authentication to identify authorized users.
- Implemented servo motor control for automatic locking and unlocking.
- Integrated a DHT22 sensor to monitor temperature and humidity.
- Used an OLED display to show container information and status

IoT-Based Waste Management and Monitoring System| Github *[C++]*

- Developed an IoT-based waste monitoring system using ultrasonic sensors to measure bin fill levels.
- Programmed ESP32 to collect sensor data and transmit it through HTTP REST API requests.
- Developed a Spring Boot backend to process incoming IoT device data.
- Integrated PostgreSQL for data storage and management.
- Developed a React dashboard to monitor bin fill levels and status.

Transport Container Security and Tracking System| Github *[C++]*

- Developed an IoT-based smart security and tracking system for transport containers.
- Implemented RFID-based authentication for electronic locking and unlocking.
- Integrated GPS tracking to obtain and transmit container location.
- Implemented GSM-based SMS notifications for location updates and lock status.
- Integrated an MPU6050 sensor to detect suspicious movement and potential tampering.
- Used an LDR sensor to detect light exposure associated with unauthorized access.

Bluetooth-Controlled Robotic Car | Github *[C++]*

- Designed and developed a Bluetooth-controlled robotic car.
- Integrated a Bluetooth module to receive wireless control commands from a smartphone.
- Implemented motor control logic to control forward, backward, left, and right movements.
- Interfaced a motor driver with DC motors to control the vehicle's movement and direction.

Feedback Analytics Module | Github *[Vue.js · Go]*

- Built a feedback analytics module with a Go backend and a Vue.js frontend, designing the request/response data flow between the service layer and the UI - directly relevant to structuring device-to-cloud and telemetry data pipelines.
- Implemented Go handlers to process, aggregate, and serve incoming feedback data through REST endpoints, applying concurrency-friendly patterns suited to handling continuous data streams.
- Presented processed results in a responsive Vue.js dashboard, giving hands-on experience turning raw backend data into a readable monitoring-style interface.

E-Commerce Site | Github *[React · JavaScript]*

- Developed a full e-commerce web application using React with product browsing, cart management, and a component-based responsive UI.
- Integrated backend data through REST-style API calls and managed client-side application state, reinforcing the request-driven communication model common in connected-device dashboards.
- Structured reusable components and handled asynchronous data fetching, error states, and loading behavior for a smooth user experience.

Event Management System (Spring Boot) | Github *[Spring Boot · Java]*

- Built an event management backend with Spring Boot (Java), implementing REST APIs, relational data models, and full CRUD operations.
- Designed the service and repository layers following an MVC structure, applying object-oriented design principles transferable to structured firmware and embedded application code.
- Modeled entity relationships and persistence with a relational database, gaining practical schema-design experience relevant to storing sensor and event data.

Pharmacy Management System | Github *[PHP · MySQL]*

- Developed a pharmacy management system using PHP and MySQL, covering inventory tracking, records management, and CRUD workflows over a relational database.
- Designed the MySQL schema and wrote queries for stock, supplier, and transaction data, building a foundation in structured data storage and retrieval.

Event Management System (PHP) | Github *[PHP · MySQL · Figma]*

- Built a PHP/MySQL event management application with database-backed record handling for events, registrations, and user data.
- Designed the interface in Figma before implementation, then translated the design into a functional front end - practicing a design-first, requirements-driven workflow.

Smart Municipal Council Page UI | Figma *[Figma · UI/UX Design]*

- Designed the user interface for a smart municipal council web page in Figma, focusing on clean layout, usability, and a consistent design system.
- Structured the information hierarchy and reusable UI components for a smart-city / connected-services concept, aligning with the kind of dashboards used to surface IoT and civic-sensor data.

CERTIFICATES

Robotics – Finalist in Vertex line following robotic challenge, Institute of Technology University of Moratuwa

Google AI Essentials - Specialization (5 Courses) - Google / Coursera (Verify: coursera.org/verify/specialization/FV0E4AZCY7BE)

Python for Beginners - Dept. of Computer Science & Engineering, University of Moratuwa (CODL) (Verify: open.uom.lk/verify · c5QrurFxnF)

VOLUNTEERING & INTERESTS

- Keen interest in IoT, connected devices, and embedded firmware development - actively self-studying microcontrollers and sensor interfacing.
- Experience in project and team management through university group projects, coordinating tasks and deliverables across team members.
- Actively building both soft and technical skills alongside the undergraduate programme.
- Strong communication and collaborative problem-solving skills.

REFERENCES

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